Exercise - 4.4

1. By the principle of mathematical induction, prove that, for $n \geq 1$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

2. By the principle of mathematical induction, prove that, for $n \geq 1$

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}.$$

3. Prove that the sum of the first n non-zero even numbers is $n^2 + n$.
4. By the principle of Mathematical induction, prove that, for $n \geq 1$

$$1.2 + 2.3 + 3.4 + \cdots + n.(n+1) = \frac{n(n+1)(n+2)}{3}.$$

5. Using the Mathematical induction, show that for any natural number $n \geq 2$,

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right)\cdots\left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}.$$

6. Using the Mathematical induction, show that for any natural number $n \geq 2$,

$$\frac{1}{1+2} + \frac{1}{1+2+3} + \frac{1}{1+2+3+4} + \cdots + \frac{1}{1+2+3+\cdots+n} = \frac{n-1}{n+1}.$$

7. Using the Mathematical induction, show that for any natural number n,

$$\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \cdots + \frac{1}{n.(n+1).(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}.$$

8. Using the Mathematical induction, show that for any natural number n,

$$\frac{1}{2.5} + \frac{1}{5.8} + \frac{1}{8.11} + \cdots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{6n+4}.$$

9. Prove by Mathematical Induction that

$$1! + (2 \times 2!) + (3 \times 3!) + ... + (n \times n!) = (n+1)! - 1.$$

10. Using the Mathematical induction, show that for any natural number n, $x^{2n} - y^{2n}$ is divisible by $x + y$.
11. By the principle of Mathematical induction, prove that, for $n \geq 1$

$$1^2 + 2^2 + 3^2 + \cdots + n^2 > \frac{n^3}{3}.$$

12. Use induction to prove that $n^3 - 7n + 3$, is divisible by 3, for all natural numbers n.
13. Use induction to prove that $5^{n+1} + 4 \times 6^n$ when divided by 20 leaves a remainder 9, for all natural numbers n.